Definite Outcomes at Infinity

The Bare Binary Biconditional, Repeated QND Measurement, and the Quantum Convergence Conjecture

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Abstract

The Bare Binary Biconditional selected $1/2$ as the unique exchange-invariant law on a bare binary alternative and connected independent repetitions of that law to half-frequency and Lebesgue measure. This paper develops the separate dynamic question that originally motivated the biconditional and the present research series: in what precise sense can an accumulating observation record approach a definite binary outcome? A self-contained classical result is given first. Let $\Theta \in \{0, 1\}$ have the symmetric prior, let observations be conditionally independent with strictly positive and distinguishable likelihoods, and let $q_n = P(\Theta = 1 \mid Y_1, \ldots, Y_n)$. Then (q_n) is a bounded martingale beginning at $1/2$, and identifiability, not martingale convergence alone, gives $q_n \to \mathbf{1}_{\{\Theta=1\}}$ almost surely. Its two endpoints therefore have equal probability. A binary-probe specialization places the observation record itself in $\{0, 1\}^{\mathbb{N}}$; a Gaussian model is retained as a continuous-observation comparison. Repeated independent runs, rather than the dependent observations within one run, produce the fair binary endpoint record connected to [1]. Bauer and Bernard's repeated quantum non-demolition measurement theorem supplies a bounded quantum counterpart: pointer weights are martingales; under their nondegeneracy condition they converge to pointer-state endpoints; the endpoint probabilities are the initial pointer weights; and relative entropy governs the asymptotic decay of losing weights. Their theorem does not choose $1/2$. This paper imports that starting value only under the explicit bare symmetric premise of [1]. Reproducible classical simulations illustrate finite-depth posterior behavior and information-rate scaling without simulating quantum dynamics or treating finite posteriors as completed limits. The remaining claim that observation gives physical significance to this convergence structure is stated separately as the author's quantum convergence conjecture.

1 Introduction and Exact Scope

The first paper in this series [1] was static. It related four descriptions of the same fair Bernoulli structure: exchange invariance of a binary law, the value $1/2$, Lebesgue measure under dyadic coding, and almost-sure half-frequency in independent records. The conjecture that motivated that paper and the present research series was subsequently described in an author op-ed [2]. It proposes that observation may make a quantum object behave as if it is at an asymptotic boundary, producing a definite binary record while unobserved behavior remains continuum-like.

The phrase *as if it is at infinity* requires a mathematical referent before it can support further research. In this paper it means one thing only:

the almost-sure limit as the number of accumulated observations $n \to \infty$.

It does not mean a last observation positioned at infinity, simultaneous occurrence of both endpoints, or a claim that every measurement mechanism needs infinitely many physical operations before producing a record. In the strictly positive comparison models below, no finite posterior equals 0 or 1 mathematically; the endpoint is a limit. More general observation channels can eliminate an alternative at a finite stage if an observed event has zero likelihood under that alternative.

This paper does not concern the quantum Zeno effect, which addresses measurement-induced inhibition rather than the convergence mechanism studied here.

Four levels are kept separate:

1. **Established classical probability:** Bayesian conditioning, martingales, identifiability, almost-sure convergence, and likelihood-ratio asymptotics.
2. **Established quantum measurement theory:** quantum filtering vocabulary and Bauer and Bernard's theorem for repeated indirect QND measurement.
3. **Synthesis and specialization:** applying the first paper's explicit bare symmetric premise to the initial two-pointer weights in the QND theorem.
4. **Author's conjecture:** the proposed physical interpretation connecting observation, asymptotic definiteness, and continuum-like unconditioned behavior.

The first two levels do not prove the fourth. The third does not add a new collapse theorem or a new probability rule. Its purpose is to show precisely where the earlier half-probability result enters an existing convergence mechanism.

No additional quantum binary measurement is proposed or required. The quantum component is the specialization of an existing repeated QND result; the numerical section supplies classical comparison figures only.

2 Four Objects That Must Not Be Confused

Let $\Theta \in \{0, 1\}$ denote the endpoint label in the classical comparison model, and let $Y_1, Y_2, ...$ be the observations collected during one run. Define

$$\mathcal{F}_n = \sigma(Y_1, ..., Y_n), \qquad \mathcal{F}_\infty = \sigma\left(\bigcup_{n\geq 1} \mathcal{F}_n\right),$$

and

$$q_n = P(\Theta = 1 \mid \mathcal{F}_n).$$

The sigma-algebra $\mathcal{F}_\infty$ is the accumulated observation sigma-algebra. It contains all information in the growing record. No tail-event theorem is needed to define it or to force the endpoint result below.

The four objects are:

In particular, a binary endpoint does not require every probe observation to be binary. Conversely, binary probe observations do not make the within-run record an independent fair Bernoulli record. The endpoint record across independently reset runs is the object that reconnects to the product measure in [1].

Table 1: Distinct objects in the convergence construction.

Object	Values	Meaning
Probe observation Y_n	Binary or continuous	One item in the within-run record
Conditional weight q_n	$[0,1]$	Probability assigned after the first n observations
Endpoint Θ and limit q_∞	$\{0,1\}$ under identifiability	Which alternative the full record identifies
Across-run endpoint record	$\{0,1\}^{\mathbb{N}}$	Outcomes from independent repetitions of the complete experiment

3 The Classical Identifiable Binary Model

3.1 Assumptions

Let

$$P(\Theta = 0) = P(\Theta = 1) = \frac{1}{2}.$$

Conditional on $\Theta = j$, suppose $Y_1, Y_2, ...$ are independent and identically distributed with density f_j relative to a common sigma-finite measure μ. Assume:

1. $f_0 > 0$ and $f_1 > 0$ μ-almost everywhere; and
2. $f_0 \neq f_1$ on a set of positive μ-measure.

Strict positivity keeps every finite posterior inside $(0,1)$. The second condition is identifiability: the two alternatives generate genuinely different observation laws.

3.2 Martingale property

Because

$$q_n = E[\mathbf{1}_{\{\Theta=1\}} \mid \mathcal{F}_n],$$

the tower property gives

$$E[q_{n+1} \mid \mathcal{F}_n] = E\Big[E[\mathbf{1}_{\{\Theta=1\}} \mid \mathcal{F}_{n+1}]\Big|\mathcal{F}_n\Big] = q_n.$$

Thus $(q_n, \mathcal{F}_n)$ is a bounded martingale. The martingale convergence theorem gives an almost-sure and L^1 limit $q_\infty \in [0,1]$ [6].

That fact alone does not imply a binary endpoint. The constant process $q_n = 1/2$ is a bounded martingale and never approaches 0 or 1. Endpoint identification requires the distinguishability assumption.

3.3 Likelihood-ratio identification

Define the likelihood ratio

$$\Lambda_n = \prod_{k=1}^{n} \frac{f_1(Y_k)}{f_0(Y_k)}.$$

Under the symmetric prior, Bayes' rule gives

$$q_n = \frac{\Lambda_n}{1+\Lambda_n}.$$

Let

$$\rho = \int \sqrt{f_0 f_1}\, d\mu$$

be the Hellinger affinity. Cauchy–Schwarz gives $\rho \leq 1$, with equality only when $f_0 = f_1$ almost everywhere. Identifiability therefore gives $\rho < 1$.

Under the law $P_0 = P(\cdot \mid \Theta = 0)$, conditional independence yields

$$E_0[\sqrt{\Lambda_n}] = \rho^n.$$

For every $\delta > 0$, Markov's inequality gives

$$P_0(\sqrt{\Lambda_n} > \delta) \leq \frac{\rho^n}{\delta}.$$

The right-hand side is summable in n. Applying Borel–Cantelli for the countable sequence $\delta = 1/m$, $m \in \mathbb{N}$, therefore implies $\Lambda_n \to 0$, P_0-almost surely. Exchanging f_0 and f_1 gives $\Lambda_n \to \infty$, P_1-almost surely. Consequently,

$$q_n \longrightarrow \mathbf{1}_{\{\Theta=1\}} \qquad \text{almost surely.}$$

This also identifies the general upward-martingale limit:

$$q_n \longrightarrow E[\mathbf{1}_{\{\Theta=1\}} \mid \mathcal{F}_\infty] = \mathbf{1}_{\{\Theta=1\}}.$$

The first equality is Lévy's upward theorem [6]; the second is the contribution of identifiability. The full accumulated record identifies the endpoint up to a null set.

3.4 Equal endpoint weights

Since $q_n \to q_\infty$ in L^1,

$$E[q_\infty] = \lim_{n\to\infty} E[q_n] = E[q_0] = \frac{1}{2}.$$

The identification step has already shown that $q_\infty \in \{0, 1\}$ almost surely. Therefore

$$P(q_\infty = 1) = E[q_\infty] = \frac{1}{2}, \qquad P(q_\infty = 0) = \frac{1}{2}.$$

The division of labor is exact:

- the symmetric prior fixes the conserved mean at $1/2$;

- the martingale property preserves that mean;
- identifiability forces the limit to the binary endpoints; and
- the binary endpoint property converts the conserved mean into endpoint probabilities.

No stopping argument is required.

4 Binary-Probe Specialization

Let the probe observations themselves be binary, $Y_n \in \{0, 1\}$, and choose an accuracy parameter $a \in (1/2, 1)$:

$$P(Y_n = 1 \mid \Theta = 1) = a, \qquad P(Y_n = 1 \mid \Theta = 0) = 1 - a.$$

If $K_n = Y_1 + \cdots + Y_n$, the log-likelihood ratio is

$$\log \Lambda_n = (2K_n - n) \log\Big(\frac{a}{1-a}\Big),$$

and

$$q_n = \Big[1 + \exp\Big(-(2K_n - n) \log\frac{a}{1-a}\Big)\Big]^{-1}.$$

The Hellinger affinity is $2\sqrt{a(1-a)} < 1$, so the theorem applies. The information separating the two conditional laws per observation is

$$D(\text{Ber}(a) \parallel \text{Ber}(1-a)) = (2a - 1) \log\Big(\frac{a}{1-a}\Big).$$

This model clarifies the connection to [1] without equating unlike records. Before conditioning on Θ, each single Y_n has marginal probability $1/2$, but the variables Y_n within one run are dependent through their shared endpoint. They do not carry the fair independent product law of [1].

Now independently reset and repeat the complete experiment. Let Θ_r be the endpoint identified in run r. Under the symmetric preparation and independent runs,

$$(\Theta_1, \Theta_2, ...) \sim \text{Ber}(1/2)^{\otimes \mathbb{N}}.$$

That across-run endpoint record is the fair product record in [1]. The earlier biconditional fixes the initial and ensemble law; the present convergence argument describes how an informative within-run record approaches one endpoint.

5 Repeated QND Measurement

5.1 The Bauer–Bernard result

Bauer and Bernard study repeated indirect quantum non-demolition measurement in a fixed pointer basis $\{|\alpha\rangle\}_{\alpha \in A}$ [3]. For a probe outcome i, let $p(i \mid \alpha)$ be the outcome probability when the system is in pointer state $|\alpha\rangle$, and let

$$q_n(\alpha) = |\langle \alpha | \varphi_n \rangle|^2$$

be the pointer weight after n probe measurements. Their update is

$$q_{n+1}(\alpha) = q_n(\alpha) \frac{p(i \mid \alpha)}{\sum_{\beta \in A} q_n(\beta) p(i \mid \beta)},$$

where the next probe outcome i has conditional probability

$$\pi_n(i) = \sum_{\beta \in A} q_n(\beta) p(i \mid \beta).$$

For every pointer label α, $q_n(\alpha)$ is a bounded martingale. Bauer and Bernard then impose the nondegeneracy condition that every pair of distinct pointer labels is distinguished by at least one probe outcome:

$$\alpha \neq \beta \quad \Longrightarrow \quad \exists i : \; p(i \mid \alpha) \neq p(i \mid \beta).$$

Under that condition, the limiting pointer-weight vector has one component equal to 1 and all others equal to 0. The probability of the endpoint γ is its initial weight $q_0(\gamma)$. They further show that, conditional on endpoint γ, a losing weight $q_n(\alpha)$ decays exponentially at a rate governed by the relative entropy between the associated probe-outcome laws [3].

These are the exact contributions used here. The result concerns the martingales formed by the pointer weights. It does not state that an arbitrary state vector is a classical martingale, and martingale convergence is not the step that independently supplies the pointer endpoints; their nondegeneracy argument does.

5.2 Bare symmetric specialization

Restrict to two pointer alternatives, labeled 0 and 1. This paper assumes the bounded bare symmetric premise stated in [1]: when the alternatives are exchange-symmetric and the admitted physical description contains no outcome-distinguishing structure within that paper's stated scope, the one-trial law is represented as exchange-invariant. That premise selects

$$q_0(0) = q_0(1) = \frac{1}{2}.$$

Applying Bauer and Bernard's theorem under their QND and nondegeneracy assumptions then gives

$$q_n(1) \longrightarrow q_\infty(1) \in \{0, 1\}$$

almost surely, with

$$P(q_\infty(1) = 1) = q_0(1) = \frac{1}{2}.$$

Bauer and Bernard do not choose the value $1/2$. Their theorem preserves the initial pointer weights and identifies them with endpoint probabilities. The value enters only through the separate symmetric premise imported from [1]. Conversely, the bare binary biconditional does not supply the QND dynamics, nondegeneracy condition, or relative-entropy rate. The two results have distinct jobs.

Quantum filtering theory supplies the broader language in which a conditional state is determined by the observation record and updated as information accumulates [4]. The present paper uses that vocabulary but does not derive a general filtering equation. Maassen and Kümmerer's purification result identifies a neighboring boundary: repeated perfect measurement either asymptotically purifies or reaches dark subspaces with unitary evolution [5]. Those exceptional and degenerate settings may motivate a separate extension, but they are not needed to repair or complete the identifiable QND case presented here.

6 Finite Posterior Simulations

6.1 What is simulated

The numerical study simulates the classical models in Sections 3 and 4. It does not simulate the system-probe unitary interaction in Bauer and Bernard's theorem. Sampling a latent endpoint Θ and then drawing from its conditional observation law is an equivalent way to sample the classical mixture distribution; it is not an assertion that a quantum measurement outcome was ontologically predetermined.

Every run begins at $q_0 = 1/2$. Using a pseudorandom generator with fixed seed `20260828`, the calculation generates 5,000 independent runs and 400 observations per run. It includes:

- binary probes with $a = 0.54$ and $a = 0.70$; and
- Gaussian observations $N(m, 1)$ under $\Theta = 1$ and $N(-m, 1)$ under $\Theta = 0$, with $m = 0.08$ and $m = 0.30$.

For the Gaussian comparison,

$$\log \Lambda_n = 2m \sum_{k=1}^{n} Y_k, \qquad D(N(m,1) \| N(-m,1)) = 2m^2.$$

The Gaussian observation record is real-valued; only its identified endpoint is binary.

6.2 Finite trajectories and distributions

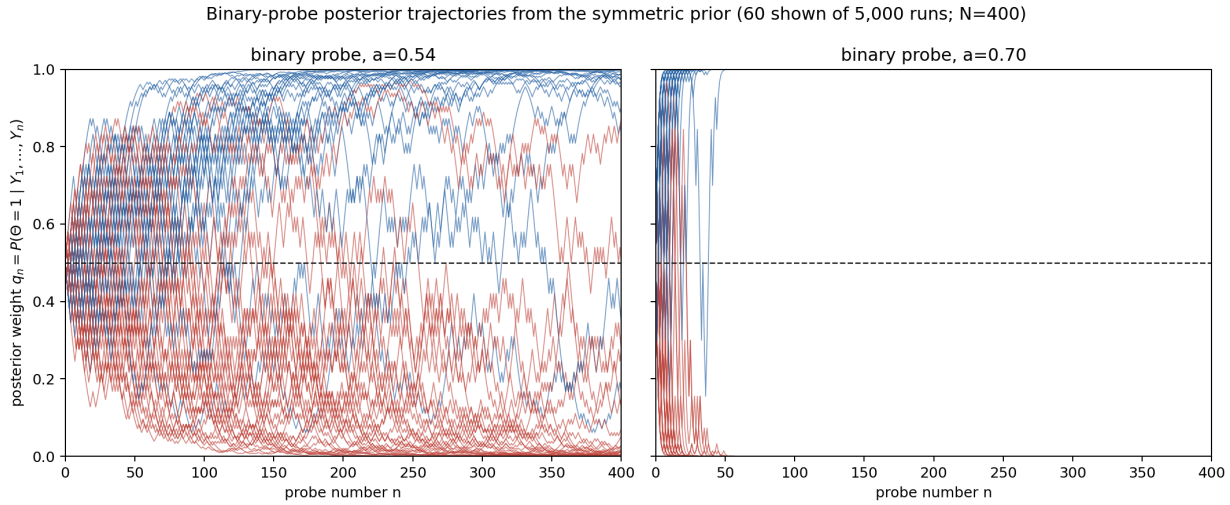


Figure 1: Sixty of 5,000 binary-probe posterior trajectories at each observation strength. Colors indicate the sampled mixture component used to generate the classical record. They do not encode an additional physical claim.

The trajectories start at $1/2$, fluctuate as observations arrive, and become increasingly concentrated near the endpoint favored by the full record. Greater distinguishability produces faster concentration.

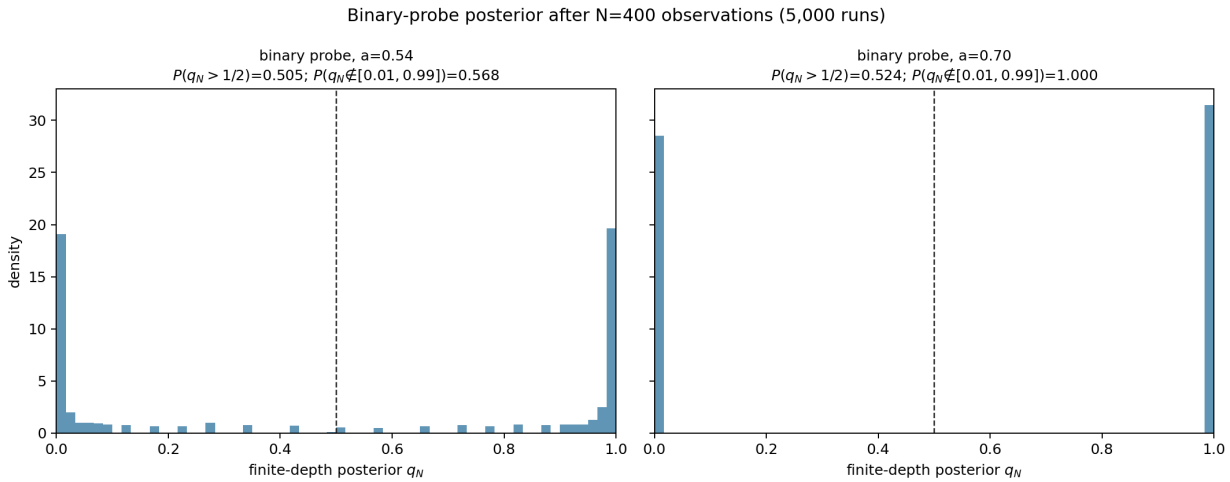


Figure 2: Distribution of q_{400} in the binary-probe simulations. Both labels are finite-depth statistics. Neither panel is called a distribution of completed limits.

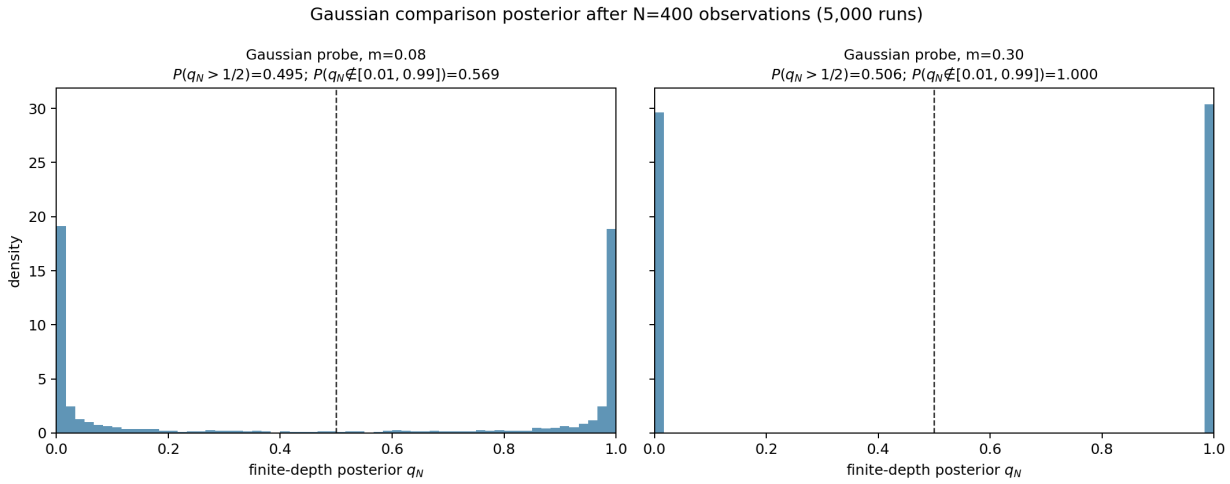


Figure 3: Distribution of q_{400} in the Gaussian comparison models, labeled with the same finite-depth quantities as Figure 2.

The numerical summaries are shown below. For the finite classification column, $q_{400} > 1/2$ predicts endpoint 1 and $q_{400} \leq 1/2$ predicts endpoint 0, so exact ties are assigned to 0.

Table 2: Finite posterior summaries from 5,000 runs at $N = 400$.

Model	**Parameter**	Endpoint fraction 1	Mean q_{400}	$P(q_{400} > 1/2)$	$P(q_{400} \notin [0.01, 0.99])$	Finite accuracy
Binary	$a = 0.54$	0.5128	0.5088	0.5050	0.5678	0.9402
Binary	$a = 0.70$	0.5242	0.5242	0.5242	1.0000	1.0000
Gaussian	$m = 0.08$	0.4918	0.4945	0.4946	0.5686	0.9460
Gaussian	$m = 0.30$	0.5062	0.5062	0.5062	1.0000	1.0000

The endpoint fractions are realized finite-sample proportions from independently sampled endpoints; no balance constraint or seed selection was imposed. At 5,000 runs, the nominal standard error of an endpoint fraction under the symmetric law is approximately 0.0071. The posterior means fluctuate around the conserved value. In exact mathematics, strict positivity keeps finite posteriors inside $(0, 1)$; strong trajectories may round to machine 0 or 1 in floating-point arithmetic.

6.3 Information-rate calibration

For the endpoint-one component of the Gaussian model, the log-likelihood ratio has mean increment $2m^2$. Reaching $q_n \geq 0.99$ is equivalent to reaching the log-odds threshold

$$b = \log\left(\frac{0.99}{0.01}\right).$$

Ignoring threshold overshoot gives the descriptive drift benchmark

$$n_{\text{benchmark}} = \frac{b}{2m^2}.$$

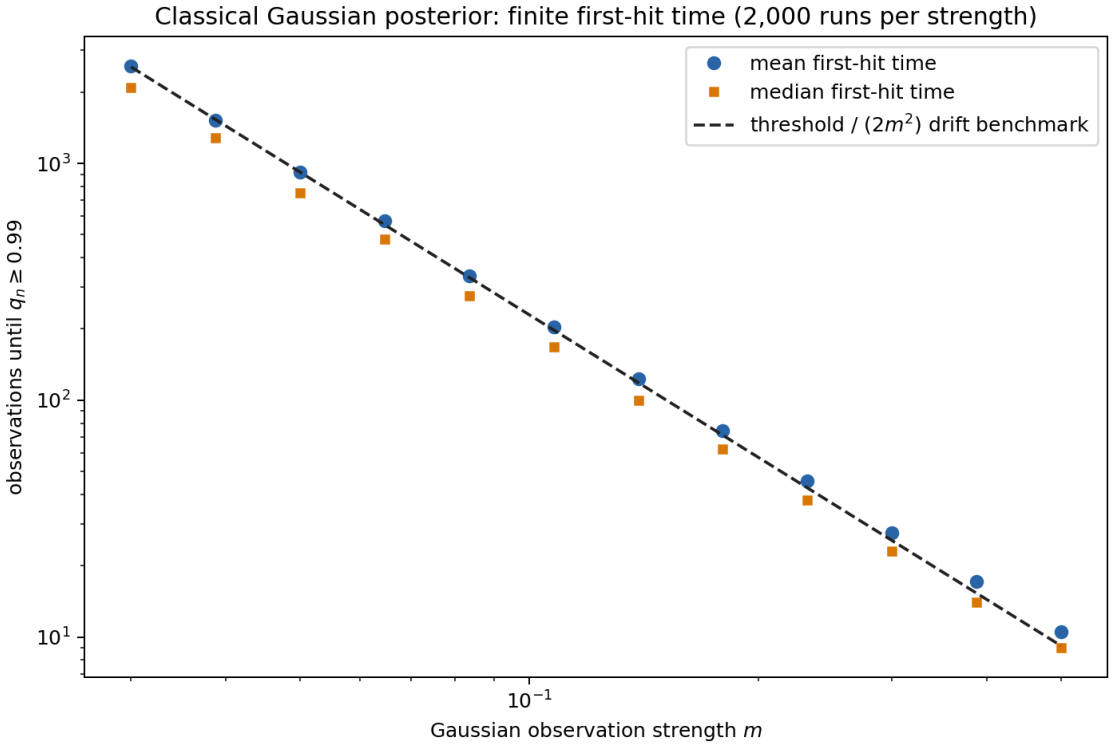


Figure 4: Mean and median finite first-hit times from 2,000 classical Gaussian trajectories per strength, compared with the threshold-over-drift benchmark. The simulation has no censored trajectories at the 60,000-step cap.

The observed mean first-hit time ranges from 2583.9 observations at $m = 0.03$ to 10.5 observations at $m = 0.5$. The benchmark ranges from 2552.8 to 9.19. This finite calculation illustrates information-rate scaling in the classical Gaussian model. It is not a numerical verification of the QND theorem. Its relevance to [3] is structural: both analyses express asymptotic separation through relative entropy of the observation laws.

7 What Is Established, Computed, Synthesized, and Conjectured

Table 3: Claim-status boundary for the paper.

Status	Statement
Established classical result	Under the stated positivity and identifiability assumptions, the binary posterior martingale converges almost surely to the endpoint indicator.
Established quantum result	In Bauer and Bernard's repeated QND setting, pointer weights are martingales; nondegeneracy forces pointer-state endpoints; endpoint probabilities equal initial weights; relative entropy governs losing-weight decay.
Computation	Binary and Gaussian classical models display finite posterior concentration and the expected information-rate scale.
Synthesis	Applying the explicit bare symmetric premise in [1] to a two-pointer initial weight gives $q_0 = (1/2, 1/2)$, which the QND theorem carries to equal endpoint probabilities.
Conjecture	The author proposes that this convergence structure may capture a physical sense in which observation produces asymptotic binary definiteness while unobserved behavior remains continuum-like.

The table prevents the simulation from becoming evidence for the theorem, the theorem from becoming a derivation of $1/2$, or the bounded synthesis from becoming proof of the broader conjecture.

8 The Remaining Quantum Convergence Conjecture

The author's motivating conjecture can now be stated more narrowly:

> ***Quantum convergence conjecture.*** *The mathematical distinction between a finite-depth conditional description and its definite asymptotic record may reflect a physical distinction between unobserved continuum-like behavior and observation-conditioned binary definiteness.*

In the motivating interpretation described in [2], *continuum-like behavior* names the wave-like and superposed behavior that an unobserved quantum object may exhibit before a definite observation record is obtained. That language states the physical possibility motivating the comparison; it does not identify the scalar posterior q_n with a quantum superposition or imply that the simulations model interference. The present paper supplies a bounded convergence analogue for the proposed transition.

The results above give this conjecture a formal comparison surface. Observation is represented by conditioning on an accumulated record; finite depth is represented by $q_n \in (0, 1)$ in the positive-likelihood models; and definite asymptotic identification is represented by $q_\infty \in \{0, 1\}$. Bauer and Bernard show that an analogous pointer-weight convergence is a theorem in their repeated QND setting.

What remains conjectural is the physical identification itself. The present paper does not prove that an unobserved object is literally a finite posterior, that measurement creates an ontological endpoint absent beforehand, or that the classical latent-variable construction is a hidden account of quantum behavior. It also does not produce a prediction that differs from the established QND result. The conjecture's value at this stage is directional: it specifies which relation between conditioned records, unconditioned evolution, and binary endpoints would need further formal or experimental development.

9 Limitations and Future Boundary

The classical theorem assumes conditionally independent identical observations with positive distinguishable densities. Non-identical designs, dependent noise, misspecification, or non-identifiable channels require separate treatment.

The binary and Gaussian figures are finite Monte Carlo illustrations with fixed parameters and seeds. The strong settings reach floating-point saturation; this is a numerical representation of concentration, not an exact finite endpoint.

The QND section is a specialization and synthesis of [3], not a derivation from first principles. The paper does not reproduce the system-probe unitary model numerically and does not extend Bauer and Bernard's nondegeneracy result.

A separate extension could study channels that fail identifiability, degenerate QND probe laws, or dark-subspace behavior. In those cases an interior or subspace-valued limit can persist because the record does not distinguish every alternative. Such an extension is not required for the identifiable binary case considered here.

10 Conclusion

The phrase *definite outcomes at infinity* has been reduced to a precise convergence statement. In the classical identifiable binary model of Section 3, the posterior weight is a bounded martingale, but its binary endpoint comes from distinguishability of the two observation laws. Starting at the bare symmetric value $1/2$, the endpoint probabilities are equal because the identified limit preserves the martingale mean.

The binary-probe specialization in Section 4 places the within-run record in a binary sequence space while showing why it must not be confused with the fair independent product record. The latter appears across independently repeated complete experiments, where the endpoint law is $\mathrm{Ber}(1/2)$. This supplies the exact connection back to the product-measure and half-frequency structure of [1].

Section 5 supplies the corresponding bounded quantum result. Bauer and Bernard's repeated-QND pointer weights are martingales; nondegeneracy produces pointer-state endpoints; initial weights determine endpoint probabilities; and relative entropy controls asymptotic separation. Their theorem supplies the convergence dynamics, while the earlier bare binary premise supplies the symmetric initial value. Neither result silently performs the other's work.

These results combine into four explicit statements:

1. Section 3 gives *as if at infinity* an exact classical referent: along an accumulating identifiable observation record, q_n converges almost surely to the endpoint indicator $q_\infty \in \{0, 1\}$.

2. Section 5.1 identifies an established quantum counterpart in the repeated-QND setting: the conditional pointer weights converge to pointer-state endpoints under Bauer and Bernard's nondegeneracy condition.

3. Section 5.2 imports the bare symmetric value $q_0 = (1/2, 1/2)$ from [1]; Bauer and Bernard's theorem preserves initial weights but does not independently select $1/2$.

4. Combining the preceding two statements carries the bare two-pointer preparation to definite QND endpoints with probability $1/2$ for each endpoint.

Section 6 illustrates the finite-depth approach to those limits with binary and Gaussian classical models. It shows conditional weights moving through the interior of $[0, 1]$, endpoint frequencies fluctuating around the conserved half-weight, and concentration time scaling with information rate. Those simulations neither create the theorems nor model quantum interference; they make the finite-to-asymptotic distinction inspectable.

Relative to the motivating op-ed [2], this paper makes two of its central phrases more specific. *Behaves as if it is at infinity* now refers to the almost-sure endpoint of an accumulating conditional record, not to a spatial location or a finite observation performed at infinity. A *stable binary state* has a bounded referent in the repeated-QND result: under nondegeneracy, the conditional pointer weights converge to a pointer-state endpoint whose probability is its initial weight. In the bare symmetric case, [1] supplies those initial weights as $1/2$ and $1/2$.

The normalization connection to [1] is exact. At binary depth n, the 2^n cylinders each have weight 2^{-n}, so their weights sum to 1; the completed fair-binary product measure also has total mass 1, and its dyadic pushforward is Lebesgue measure on $[0, 1]$. In the two-pointer QND specialization, the conditional weights likewise sum to 1 at every stage, beginning at $(1/2, 1/2)$ and converging to either $(1, 0)$ or $(0, 1)$. Thus $1/2$ is the equal weight of each bare binary alternative, while 1 is the total weight of the exhaustive alternatives.

In the physical comparison proposed in [2], a normalized wave function ψ over its full domain Ω satisfies the corresponding unit-total relation

$$\int_{\Omega} |\psi(x)|^2 \, dx = 1.$$

A definite particle-like observation does not, by itself, imply that the wave description has ceased to apply. Taken together, the normalization and convergence results supply the intended mathematical correspondence: unit total weight belongs to the full wave superposition, while repeated conditioning on observation produces a definite endpoint record. In the conjecture's reading, the wave's appearance as a particle is therefore the expected observational form of the full normalized superposition, not evidence that the object has ceased to be a wave. Establishing this correspondence mathematically is the aim of the present synthesis.

The mathematics is clear for the repeated-QND setting and its bare symmetric specialization: the pointer weights converge to definite endpoints, and each endpoint has probability $1/2$. What remains untested here is whether this mathematical correspondence is the exact physical mechanism realized in observation. The paper makes no claim to that physical demonstration.

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Preparation Note

Fable 5 suggested repeated-measurement convergence as a potential way to test the author's pre-existing quantum convergence conjecture, which had motivated the bare binary biconditional. The author directed the sampling study and subsequently directed the scope and corrections of this paper. GPT 5.6 assisted with source audit, probability-terminology review, classical proof development, computation audit, figure preparation, and drafting. The author retains responsibility for the claims, interpretations, and final text.